RESEARCH STATEMENT

Machine learning for materials discovery, checked against reality

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The question

Machine learning can now predict a material's properties in milliseconds, and screening hundreds of thousands of hypothetical compounds has become routine. But a prediction helps discovery only if we know when to trust it, and the materials we most want, with new chemistry or unusual properties, are exactly where models have seen the least data. One question runs through all my work: *when an ML prediction replaces an expensive calculation or a measurement, does the conclusion survive the check?* I study it in thermoelectrics, where the key properties, the Seebeck coefficient and the lattice thermal conductivity (κ_L), are costly to compute and hard to measure, and where the Heusler family offers a large, chemically rich space to search.

What I have learned so far

1. **Accuracy on familiar data is not discovery.** In my Summer Research Fellowship at the S. N. Bose Centre (2025), graph neural networks trained on narrow perovskite families scored above 92% on familiar data but about 48% on unseen double perovskites, while a broadly pretrained model kept about 80%. The breadth of the training data, not the test score, decided whether a model generalized.
2. **A cheap ML signal can choose what to calculate, but not what is true.** I trained 30 ALIGNN models to predict the density of states (DOS) of 54,978 Heusler structures and judged a DOS-based screen against 3,625 first-principles transport calculations from my own Quantum ESPRESSO + BoltzTraP2 workflow [P1]. The screen needed about four times fewer calculations per confirmed candidate and gave 80 candidate thermoelectrics, 61 never measured. Yet the predicted spectra missed most band gaps (recall 0.15) despite a held-out correlation of $r = 0.90$: standard metrics hid the failure that mattered most for the physics.
3. **Models inherit the bias of the data they learn from.** I showed that published calculations overestimate the measured κ_L of half-Heuslers by a median 1.69× [P2]. I built a source-tracked dataset of 9,495 values for 960 compounds, partly extracted from the literature by multimodal language models with every value physics-checked, and a two-parameter correction from calculation to experiment. For cubic 18-electron compounds, ML predictions of measured κ_L reached a 33% median error against 46–47% for baselines, close to the disagreement between two laboratories.

Both studies are first-author papers, now submitted, and their code and data are public.

Now

At NIMS, Japan, in the Spin Theory Group of Prof. Terumasa Tadano, I am asking whether universal machine-learned interatomic potentials, pretrained on millions of DFT calculations, already encode what is needed to predict transport. Using frozen transfer learning, I am applying them to new alkali- and alkaline-earth-metal Heusler compositions built combinatorially, and ranking these by predicted Seebeck coefficient and lattice thermal conductivity.

What I want to do in a PhD

1. **Foundation models beyond energies and forces.** How far do the representations learned by universal potentials transfer to phonon, thermal and electronic-transport properties, and how can we tell when they do not?
2. **Uncertainty that measures discovery, not interpolation.** Prediction intervals and benchmarks built on genuinely new chemistries and structure types, so that a screening campaign can state how many of its candidates will survive DFT and experiment.
3. **Closing the loop.** Property-conditioned generative models propose crystals, calibrated surrogates and first-principles calculations filter them, and measurements correct the surrogates. This is the loop that autonomous laboratories depend on, and it works only if every step can be trusted.

What I bring

I have worked across the whole chain: curating data from databases and the literature, training graph neural networks and interatomic potentials on GPU clusters, running high-throughput DFT and transport workflows, and comparing the results with experiment. I release the code and data with every project, and I have mentored two junior students who are co-authors on [P2]. I am looking for a group that combines machine learning with first-principles theory or experiment, where a prediction counts when it leads to a real material.